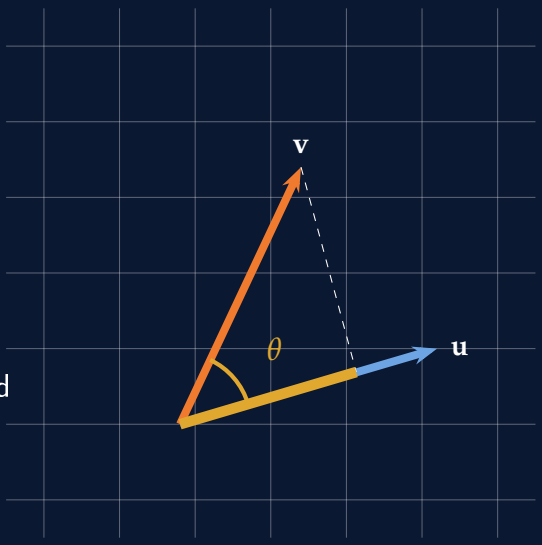
ZMATHLAB • LINEAR ALGEBRA

Norm, Dot Product and Distance

How big? How far? Which way? The three questions behind every vector



Lecture notes for BS students

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Based on: H. Anton, C. Rorres & A. Kaul, *Elementary Linear Algebra*, Section 3.2

What you will be able to do after these notes

- find the **norm** (length) of any vector in $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{R}^n$, and explain it with Pythagoras;
- build **unit vectors** and write vectors with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ and $\mathbf{e}_1, \ldots, \mathbf{e}_n$;
- compute the **distance** between points in any dimension;
- use the **dot product** both ways (lengths and angle, or components) and read the angle from its sign;
- use Cauchy–Schwarz, the triangle inequality and the parallelogram equation;
- avoid the classic mistakes and solve every problem of Exercise Set 3.2.

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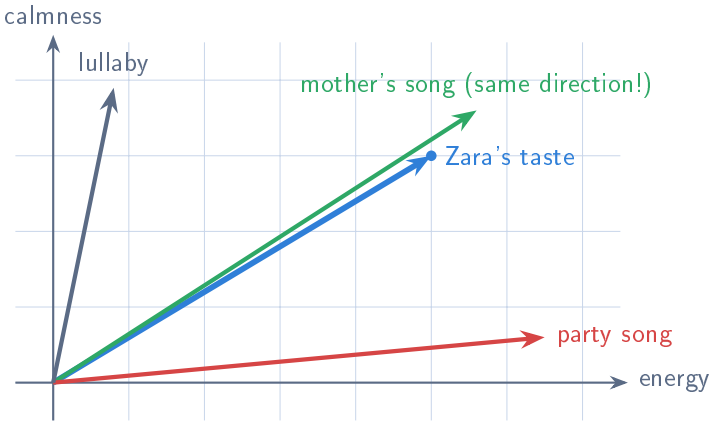
1 The story: the song nobody searched for

Story

Zara's first night in a university hostel. A new city, nobody she knows. She puts on her headphones and presses *shuffle*.
The app plays an old song, the exact song her mother used to hum in the kitchen every evening. Zara never searched for it. Not once.
How did a phone know? It has never met her mother. It has never seen that kitchen.
The secret: to the app every song is a **list of numbers** (tempo, mood, energy, rhythm, ...), a **vector**. Zara's taste is a vector too. To find the song, the phone asked three questions:

How big is a vector? **How far apart are two vectors?** **Do two vectors point the same way?**

The answers have names: the **norm**, the **distance** and the **dot product**.



Small angle = similar taste. The dot product measures exactly this.

2 Why these three ideas? Applications

Maps and ride apps (distance)

Which driver is nearest? The app computes the distance $d = \|P_1 - P_2\|$ from you to every driver and picks the smallest.

Music, Netflix, YouTube (dot product)

Recommendation systems compare taste vectors with $\cos\theta = \dfrac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$ ("cosine similarity"): close to 1 means "very similar".

Physics: work done

Pushing a cart with force $\mathbf{F}$ through displacement $\mathbf{d}$ does work $W = \mathbf{F}\cdot\mathbf{d}$. Only the part of the force *along* the motion counts.

Computer graphics: light and shade

A surface is bright when it faces the light: brightness $\propto \mathbf{n}\cdot\mathbf{L}$ (normal · light direction). Negative means "in shadow".

Search engines and AI

Documents and words are vectors in $\mathbb{R}^n$ with huge n. "Similar meaning" = small an-

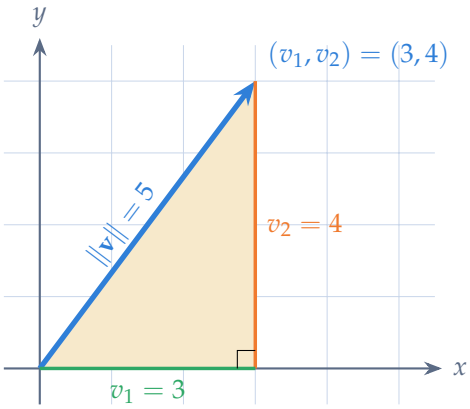
gle between the vectors.

ISBN check digits

The 10th digit of an old ISBN is $(1, 2, \ldots, 9) \cdot (\text{first nine digits}) \bmod 11$; it catches typing errors (Section 13).

3 The norm: how long is an arrow?

3.1 In 2-space: one Pythagoras

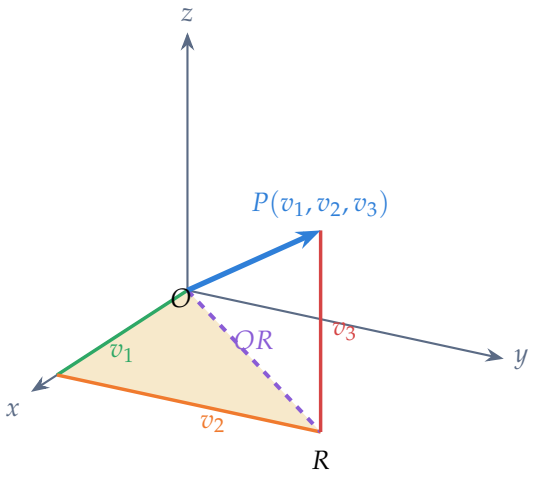


By the theorem of Pythagoras, the length of $\mathbf{v} = (v_1, v_2)$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2} \tag{1}$$

For $(3, 4)$: $\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

3.2 In 3-space: Pythagoras twice



First on the floor: $(OR)^2 = v_1^2 + v_2^2$. Then up: $\|\mathbf{v}\|^2 = (OR)^2 + (RP)^2 = v_1^2 + v_2^2 + v_3^2$. So

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2} \tag{2}$$

3.3 In n-space: the same pattern

Definition: Norm (length, magnitude) – Anton Definition 1

If $\mathbf{v} = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n$, the **norm** of $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \tag{3}$$

In simple words

Square every component, **add** them all, take the **square root**. That's it, in any dimension.

Example: Anton, Example 1 (every step)

$\mathbf{v} = (-3, 2, 1)$: $\|\mathbf{v}\| = \sqrt{(-3)^2 + 2^2 + 1^2} = \sqrt{9 + 4 + 1} = \sqrt{14} \approx 3.742$.
$\mathbf{v} = (2, -1, 3, -5) \in \mathbb{R}^4$: $\|\mathbf{v}\| = \sqrt{2^2 + (-1)^2 + 3^2 + (-5)^2} = \sqrt{4 + 1 + 9 + 25} = \sqrt{39} \approx 6.245$.

Theorem 3.2.1

For $\mathbf{v} \in \mathbb{R}^n$ and any scalar k: (a) $\|\mathbf{v}\| \geq 0$ (b) $\|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}$ (c) $\|k\mathbf{v}\| = |k|\,\|\mathbf{v}\|$.

Proof of (c). $k\mathbf{v} = (kv_1, \ldots, kv_n)$, so

$$\begin{aligned} \|k\mathbf{v}\| &= \sqrt{(kv_1)^2 + \cdots + (kv_n)^2} && \text{[definition of norm]} \\ &= \sqrt{k^2(v_1^2 + \cdots + v_n^2)} && \text{[factor out } k^2\text{]} \\ &= \sqrt{k^2}\,\sqrt{v_1^2 + \cdots + v_n^2} = |k|\,\|\mathbf{v}\| && \text{[}\sqrt{k^2} = |k|\text{, not } k\text{!]} \quad \blacksquare \end{aligned}$$

Why (a) and (b): a sum of squares is never negative, and it is zero only when every square is zero, i.e. every $v_i = 0$.

4 Unit vectors

Two nonzero vectors have the **same direction** if each is a positive multiple of the other, and **opposite directions** if each is a negative multiple. Example: $(2, -4, 1, 8)$ and $(1, -2, \frac{1}{2}, 4)$ have the same direction; $(2, -4, 1, 8)$ and $(-1, 2, -\frac{1}{2}, -4)$ opposite directions.

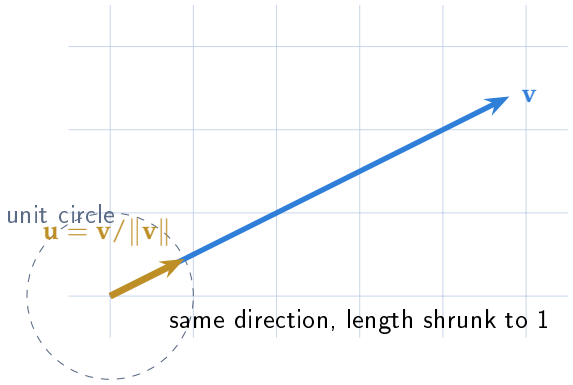
Definition: Unit vector, normalizing

A vector of norm 1 is a **unit vector**. For any $\mathbf{v} \neq \mathbf{0}$,

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|}\,\mathbf{v} \tag{4}$$

is the unit vector in the direction of $\mathbf{v}$. Producing it is called **normalizing** $\mathbf{v}$.

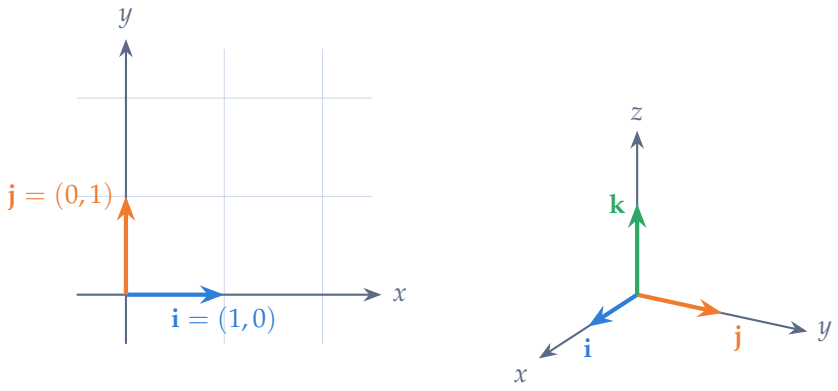
Why $\|\mathbf{u}\| = 1$: by Theorem 3.2.1(c) with $k = 1/\|\mathbf{v}\| > 0$: $\|\mathbf{u}\| = |k|\|\mathbf{v}\| = \frac{1}{\|\mathbf{v}\|}\|\mathbf{v}\| = 1$.



Example: Anton, Example 2 (every step)

$\mathbf{v} = (2,2,-1)$. $\|\mathbf{v}\| = \sqrt{2^2+2^2+(-1)^2} = \sqrt{4+4+1} = \sqrt{9} = 3$.
$\mathbf{u} = \frac{1}{3}(2,2,-1) = (\frac{2}{3},\frac{2}{3},-\frac{1}{3})$. Check: $\|\mathbf{u}\|^2 = \frac{4}{9}+\frac{4}{9}+\frac{1}{9} = \frac{9}{9} = 1$ ✓.

4.1 Standard unit vectors



In $\mathbb{R}^2$: $\mathbf{i} = (1,0)$, $\mathbf{j} = (0,1)$. In $\mathbb{R}^3$: $\mathbf{i} = (1,0,0)$, $\mathbf{j} = (0,1,0)$, $\mathbf{k} = (0,0,1)$. Every vector is a linear combination of them:

$$(v_1,v_2) = v_1\mathbf{i} + v_2\mathbf{j}, \qquad (v_1,v_2,v_3) = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}. \tag{5–6}$$

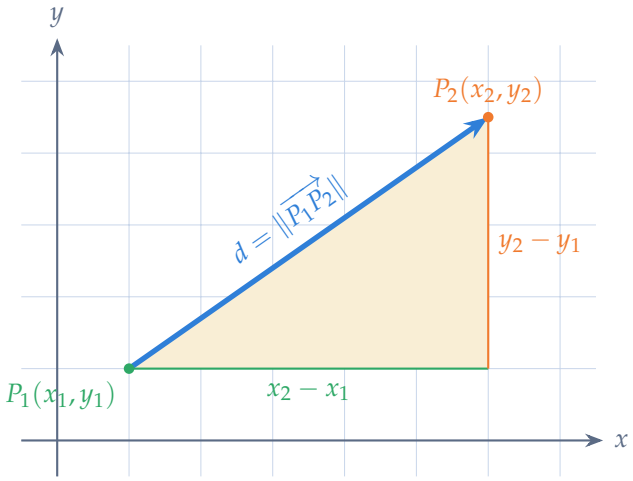
In $\mathbb{R}^n$: $\mathbf{e}_1 = (1,0,\ldots,0)$, $\ldots$, $\mathbf{e}_n = (0,\ldots,0,1)$ and $\mathbf{v} = v_1\mathbf{e}_1 + \cdots + v_n\mathbf{e}_n$.

Example: Anton, Example 3

$(2,-3,4) = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$, $\quad (7,3,-4,5) = 7\mathbf{e}_1 + 3\mathbf{e}_2 - 4\mathbf{e}_3 + 5\mathbf{e}_4$.

5 Distance

The distance between two points is the length of the arrow joining them:



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (9) \qquad d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (10)$$

Definition: Distance in $\mathbb{R}^n$ – Anton Definition 2

For points $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$: $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2}$. (11)

Example: Anton, Example 4 (every step)

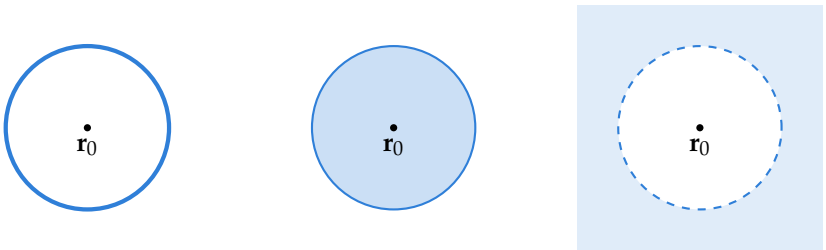
$\mathbf{u} = (1, 3, -2, 7)$, $\mathbf{v} = (0, 7, 2, 2)$. Differences: $1 - 0 = 1$, $3 - 7 = -4$, $-2 - 2 = -4$, $7 - 2 = 5$. Squares: $1,\ 16,\ 16,\ 25$; sum $= 58$. So $d(\mathbf{u}, \mathbf{v}) = \sqrt{58} \approx 7.616$.

Teacher's tip

$d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$ because $(u_i - v_i)^2 = (v_i - u_i)^2$: the order of subtraction does not matter for distance (it *does* matter for the vector $\overrightarrow{P_1P_2}$).

5.1 Mind-blowing geometry: circles and spheres are distance sets

The set of all $\mathbf{r}$ with $\|\mathbf{r} - \mathbf{r}_0\| = 1$ is the **circle** (in $\mathbb{R}^2$) or **sphere** (in $\mathbb{R}^3$) of radius 1 centred at $\mathbf{r}_0$; $\|\mathbf{r} - \mathbf{r}_0\| \leq 1$ is the filled disc / ball; > 1 is everything outside.



$\|\mathbf{r} - \mathbf{r}_0\| = 1$ $\|\mathbf{r} - \mathbf{r}_0\| \leq 1$ $\|\mathbf{r} - \mathbf{r}_0\| > 1$

6 The dot product: do two vectors agree?

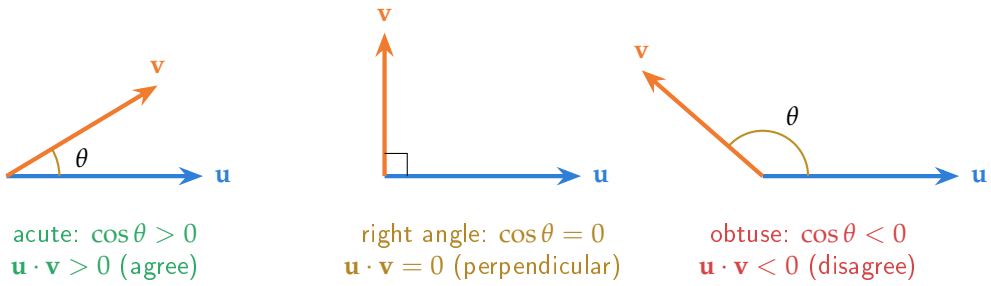
Place $\mathbf{u}, \mathbf{v}$ tail to tail. The **angle** θ between them satisfies $0 \leq \theta \leq \pi$.

Definition: Dot product, geometric form – Anton Definition 3

For nonzero $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$ with angle θ:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta \tag{12}$$

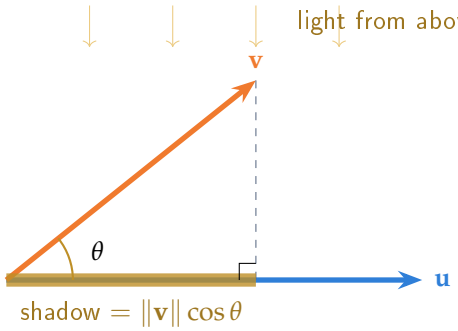
If $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$, define $\mathbf{u} \cdot \mathbf{v} = 0$. The answer is a **number** (a scalar), not a vector.



6.1 Mind-blowing geometry: the dot product is a shadow

Shine a light straight down onto $\mathbf{u}$. The shadow of $\mathbf{v}$ on the line of $\mathbf{u}$ has (signed) length $\|\mathbf{v}\| \cos\theta$. So

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \times (\text{signed shadow of } \mathbf{v} \text{ on } \mathbf{u}).$$



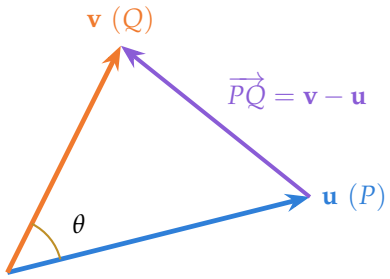
In simple words

The dot product asks: *how much of* $\mathbf{v}$ *goes in the direction of* $\mathbf{u}$*?* Big positive = strongly agree, zero = completely independent (perpendicular), negative = pull against each other.

Example: Anton, Example 5 (every step)

$\mathbf{u} = (0,0,1)$, $\mathbf{v} = (0,2,2)$, angle $45°$. $\|\mathbf{u}\| = 1$, $\|\mathbf{v}\| = \sqrt{0+4+4} = \sqrt{8} = 2\sqrt{2}$, $\cos 45° = \frac{1}{\sqrt{2}}$.
$\mathbf{u} \cdot \mathbf{v} = (1)(2\sqrt{2})\left(\frac{1}{\sqrt{2}}\right) = 2$.

7 The component formula (no angles needed)



Derivation (every step). Law of cosines in the triangle:

$$\begin{aligned}
\|\mathbf{v}-\mathbf{u}\|^2 &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta && \text{[law of cosines]}\\
\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta &= \tfrac{1}{2}\left(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - \|\mathbf{v}-\mathbf{u}\|^2\right) && \text{[rearrange]}\\
\mathbf{u}\cdot\mathbf{v} &= \tfrac{1}{2}\Big[(u_1^2+u_2^2+u_3^2) + (v_1^2+v_2^2+v_3^2) \\
&\qquad - \left((v_1-u_1)^2 + (v_2-u_2)^2 + (v_3-u_3)^2\right)\Big] && \text{[norms in components]}\\
&= \tfrac{1}{2}\left[2u_1v_1 + 2u_2v_2 + 2u_3v_3\right] && [(v_i-u_i)^2 = v_i^2 - 2u_iv_i + u_i^2]\\
&= u_1v_1 + u_2v_2 + u_3v_3. && (15)
\end{aligned}$$

Definition: Dot product in $\mathbb{R}^n$ – Anton Definition 4

$\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n.$ (17)
In words: **multiply matching components and add the products.**

Example: Anton, Example 6 (every step)

(a) $\mathbf{u} = (0,0,1)$, $\mathbf{v} = (0,2,2)$: $(0)(0) + (0)(2) + (1)(2) = 2$, the same as Example 5 with no angle at all.
(b) $\mathbf{u} = (-1,3,5,7)$, $\mathbf{v} = (-3,-4,1,0)$: $(-1)(-3) + (3)(-4) + (5)(1) + (7)(0) = 3 - 12 + 5 + 0 = -4$. Negative, so the angle is obtuse.

Teacher's tip

History. The dot notation was introduced by the American physicist J. Willard Gibbs (1839–1903) in notes for his Yale students in the 1880s.

8 Angles from the dot product

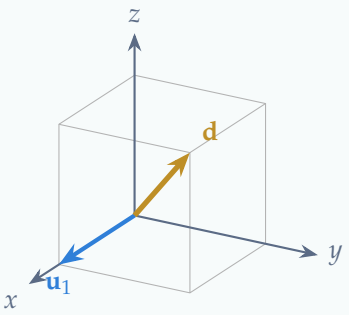
$$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} \qquad \begin{cases} \mathbf{u}\cdot\mathbf{v} > 0 & \Rightarrow\ \theta \text{ acute} \\ \mathbf{u}\cdot\mathbf{v} = 0 & \Rightarrow\ \theta = \pi/2 \\ \mathbf{u}\cdot\mathbf{v} < 0 & \Rightarrow\ \theta \text{ obtuse} \end{cases} \tag{13}$$

Example: Anton, Example 7: diagonal of a cube and an edge

Edge length k. Edges $\mathbf{u}_1 = (k,0,0)$, $\mathbf{u}_2 = (0,k,0)$, $\mathbf{u}_3 = (0,0,k)$; diagonal $\mathbf{d} = (k,k,k) = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3$.

$$\mathbf{u}_1 \cdot \mathbf{d} = k \cdot k + 0 \cdot k + 0 \cdot k = k^2$$
$$\|\mathbf{u}_1\| = k, \qquad \|\mathbf{d}\| = \sqrt{3k^2} = \sqrt{3}\,k$$
$$\cos\theta = \frac{k^2}{(k)(\sqrt{3}k)} = \frac{1}{\sqrt{3}}$$
$$\theta = \cos^{-1}\left(\tfrac{1}{\sqrt{3}}\right) \approx 54.74^\circ$$

The size k cancels: every cube has the same shape, so the angle cannot depend on its size.



9 Algebraic properties

Dotting a vector with itself gives its length squared:

$$\mathbf{v} \cdot \mathbf{v} = v_1^2 + \cdots + v_n^2 = \|\mathbf{v}\|^2, \qquad \|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}. \tag{18–19}$$

Theorem 3.2.2

For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and scalar k:
(a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ [symmetry]
(b) $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ [distributive]
(c) $k(\mathbf{u} \cdot \mathbf{v}) = (k\mathbf{u}) \cdot \mathbf{v}$ [homogeneity]
(d) $\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0 \iff \mathbf{v} = \mathbf{0}$ [positivity]

Proof of (c): $k(\mathbf{u} \cdot \mathbf{v}) = k(u_1v_1 + \cdots + u_nv_n) = (ku_1)v_1 + \cdots + (ku_n)v_n = (k\mathbf{u}) \cdot \mathbf{v}$.
Proof of (d): $\mathbf{v} \cdot \mathbf{v} = v_1^2 + \cdots + v_n^2 = \|\mathbf{v}\|^2$, then use Theorem 3.2.1(a),(b). ■

Theorem 3.2.3

(a) $\mathbf{0} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{0} = 0$ (b) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$ (c) $\mathbf{u} \cdot (\mathbf{v} - \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{w}$
(d) $(\mathbf{u} - \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} - \mathbf{v} \cdot \mathbf{w}$ (e) $k(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (k\mathbf{v})$

Proof of (b) without components: $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{w} \cdot (\mathbf{u} + \mathbf{v})$ [symmetry] $= \mathbf{w} \cdot \mathbf{u} + \mathbf{w} \cdot \mathbf{v}$ [distributive] $= \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$ [symmetry].

Example: Anton, Example 8 (expand like brackets)

$$\begin{aligned}(\mathbf{u} - 2\mathbf{v}) \cdot (3\mathbf{u} + 4\mathbf{v}) &= \mathbf{u} \cdot (3\mathbf{u} + 4\mathbf{v}) - 2\mathbf{v} \cdot (3\mathbf{u} + 4\mathbf{v}) \\ &= 3(\mathbf{u} \cdot \mathbf{u}) + 4(\mathbf{u} \cdot \mathbf{v}) - 6(\mathbf{v} \cdot \mathbf{u}) - 8(\mathbf{v} \cdot \mathbf{v}) \\ &= 3\|\mathbf{u}\|^2 + (4-6)(\mathbf{u} \cdot \mathbf{v}) - 8\|\mathbf{v}\|^2 = 3\|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) - 8\|\mathbf{v}\|^2.\end{aligned}$$

10 Cauchy–Schwarz and angles in $\mathbb{R}^n$

We want to *define* the angle in $\mathbb{R}^n$ by $\theta = \cos^{-1}\left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right)$ (20). This only works if $-1 \le \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} \le 1$ (21). It always is:

Theorem 3.2.4 (Cauchy–Schwarz inequality)

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$: $|\mathbf{u}\cdot\mathbf{v}| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$ (22)
In components: $|u_1v_1 + \cdots + u_nv_n| \le (u_1^2 + \cdots + u_n^2)^{1/2}(v_1^2 + \cdots + v_n^2)^{1/2}$. (23)

A short proof (for the curious). If $\mathbf{u} = \mathbf{0}$ both sides are 0. Otherwise, for every real t, $0 \le \|t\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2t^2 + 2(\mathbf{u}\cdot\mathbf{v})t + \|\mathbf{v}\|^2$. A quadratic in t that is never negative has discriminant ≤ 0: $4(\mathbf{u}\cdot\mathbf{v})^2 - 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2 \le 0$, i.e. $|\mathbf{u}\cdot\mathbf{v}| \le \|\mathbf{u}\|\|\mathbf{v}\|$. ■

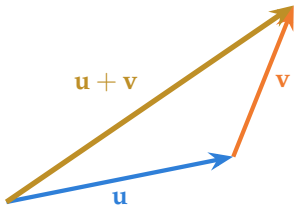
Teacher's tip

Named after Augustin Cauchy and Hermann Schwarz; the Russian Viktor Bunyakovsky published a version in 1859, about 25 years before Schwarz.

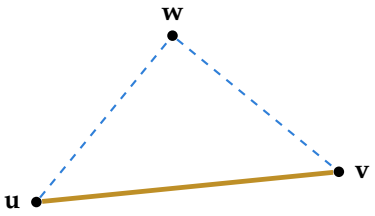
11 Geometry in $\mathbb{R}^n$

Theorem 3.2.5

(a) $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$ [triangle inequality for vectors]
(b) $d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$ [triangle inequality for distances]



straight is never longer than the detour



$d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$

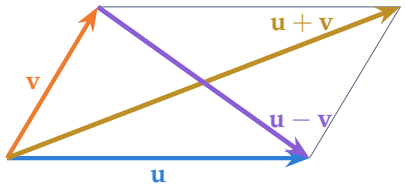
Proof of (a), every step:

$$\begin{aligned}
\|\mathbf{u}+\mathbf{v}\|^2 &= (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) = \mathbf{u}\cdot\mathbf{u} + 2(\mathbf{u}\cdot\mathbf{v}) + \mathbf{v}\cdot\mathbf{v} = \|\mathbf{u}\|^2 + 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|^2 \\
&\le \|\mathbf{u}\|^2 + 2|\mathbf{u}\cdot\mathbf{v}| + \|\mathbf{v}\|^2 && [x \le |x|] \\
&\le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 && [\text{Cauchy–Schwarz}] \\
&= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2 && [\text{perfect square}]
\end{aligned}$$

Both sides are nonnegative, so taking square roots gives (a). For (b): $d(\mathbf{u}, \mathbf{v}) = \|(\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})\| \le \|\mathbf{u} - \mathbf{w}\| + \|\mathbf{w} - \mathbf{v}\|$.

Theorem 3.2.6 (parallelogram equation)

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\left(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\right) \quad (24)$$



sum of squares of the 2 diagonals
= sum of squares of the 4 sides

Proof: $\|\mathbf{u}\pm\mathbf{v}\|^2 = \|\mathbf{u}\|^2 \pm 2(\mathbf{u}\cdot\mathbf{v}) + \|\mathbf{v}\|^2$; adding, the $\pm 2(\mathbf{u}\cdot\mathbf{v})$ terms cancel, leaving $2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$.

Theorem 3.2.7 (dot product from norms)

$$\mathbf{u}\cdot\mathbf{v} = \tfrac{1}{4}\|\mathbf{u}+\mathbf{v}\|^2 - \tfrac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2 \tag{25}$$

Proof: subtract the two expansions above: $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4(\mathbf{u}\cdot\mathbf{v})$, then divide by 4.

12 Dot products as matrix multiplication

Write $\mathbf{u}, \mathbf{v}$ as column matrices: $\mathbf{u}\cdot\mathbf{v} = \mathbf{u}^T\mathbf{v} = \mathbf{v}^T\mathbf{u}$. Example:

$$\mathbf{u} = \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix}, \qquad \mathbf{u}^T\mathbf{v} = \begin{bmatrix} 1 & -3 & 5 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix} = 5 - 12 + 0 = -7.$$

For an $n \times n$ matrix A:

$$A\mathbf{u}\cdot\mathbf{v} = \mathbf{u}\cdot A^T\mathbf{v} \quad (26) \qquad \mathbf{u}\cdot A\mathbf{v} = A^T\mathbf{u}\cdot\mathbf{v} \quad (27)$$

because $A\mathbf{u}\cdot\mathbf{v} = \mathbf{v}^T(A\mathbf{u}) = (\mathbf{v}^TA)\mathbf{u} = (A^T\mathbf{v})^T\mathbf{u} = \mathbf{u}\cdot A^T\mathbf{v}$.

Example: Anton, Example 9 (every step)

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & 1 \\ -1 & 0 & 1 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix}.$$

$$A\mathbf{u} = \begin{bmatrix} -1-4+12 \\ -2+8+4 \\ 1+0+4 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \\ 5 \end{bmatrix}, \quad A^T\mathbf{v} = \begin{bmatrix} 1 & 2 & -1 \\ -2 & 4 & 0 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} -2+0-5 \\ 4+0+0 \\ -6+0+5 \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ -1 \end{bmatrix}.$$

$$A\mathbf{u}\cdot\mathbf{v} = 7(-2) + 10(0) + 5(5) = 11, \quad \mathbf{u}\cdot A^T\mathbf{v} = (-1)(-7) + 2(4) + 4(-1) = 11 \checkmark.$$

Matrix product = table of dot products. If A has rows $\mathbf{r}_1, \ldots, \mathbf{r}_m$ and B has columns $\mathbf{c}_1, \ldots, \mathbf{c}_n$, then the (i, j) entry of AB is $\mathbf{r}_i \cdot \mathbf{c}_j$ (28).

13 Application: ISBN check digits

Old 10-digit ISBNs end in a check digit c. With $\mathbf{a} = (1, 2, \ldots, 9)$ and $\mathbf{b}$ = first nine digits: form $\mathbf{a}\cdot\mathbf{b}$, divide by 11; the remainder is c ($c = 10$ is written X).

Example: ISBN 0-471-15307-9

$\mathbf{a}\cdot\mathbf{b} = (1,2,3,4,5,6,7,8,9)\cdot(0,4,7,1,1,5,3,0,7) = 0+8+21+4+5+30+21+0+63 = 152.$
$152 = 11 \times 13 + 9$, so $c = 9$ ✓, matching the last digit.

14 Direction angles (from the exercises)

The angles α, β, γ between $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are its **direction angles**. From (13), e.g. $\cos\alpha = \frac{\mathbf{v}\cdot\mathbf{i}}{\|\mathbf{v}\|\|\mathbf{i}\|} = \frac{v_1}{\|\mathbf{v}\|}$, so

$$\cos\alpha = \frac{v_1}{\|\mathbf{v}\|}, \quad \cos\beta = \frac{v_2}{\|\mathbf{v}\|}, \quad \cos\gamma = \frac{v_3}{\|\mathbf{v}\|}, \qquad \cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{v_1^2 + v_2^2 + v_3^2}{\|\mathbf{v}\|^2} = 1.$$

These cosines are exactly the components of the unit vector $\mathbf{v}/\|\mathbf{v}\|$.

15 Common student misconceptions

Misconception 1: "the norm of a sum is the sum of the norms"

Wrong: $\|\mathbf{u}+\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$.
Right: only $\leq$ in general. $(3,0)+(0,4) = (3,4)$: norms $3+4=7$ but $\|(3,4)\| = 5$. Equality holds only when one vector is a nonnegative multiple of the other (Ex. 30).

Misconception 2: forgetting the square root

Wrong: $\|(3,4)\| = 3^2 + 4^2 = 25$. Right: $\|(3,4)\| = \sqrt{25} = 5$; the 25 is $\|\mathbf{v}\|^2$.

Misconception 3: negative scalars inside the norm

Wrong: $\|-3\mathbf{v}\| = -3\|\mathbf{v}\|$. Right: $\|-3\mathbf{v}\| = |-3|\,\|\mathbf{v}\| = 3\|\mathbf{v}\|$. A length is never negative.

Misconception 4: squaring negatives

Wrong: $\|(-3,2,1)\| = \sqrt{-9+4+1}$. Right: $(-3)^2 = +9$, so $\sqrt{9+4+1} = \sqrt{14}$. Always use brackets.

Misconception 5: the dot product is a vector

Wrong: $(1,2)\cdot(3,4) = (3,8)$. Right: it is the **number** $3+8=11$. So $\mathbf{u}\cdot(\mathbf{v}\cdot\mathbf{w})$ and $\|\mathbf{u}\cdot\mathbf{v}\|$ are meaningless (Ex. 15).

Misconception 6: $\|k\mathbf{v}\| = a$ has one answer

Wrong: $k = a/\|\mathbf{v}\|$ only. Right: $|k| = a/\|\mathbf{v}\|$, so $k = \pm a/\|\mathbf{v}\|$: two answers (Ex. 7, 8).

Misconception 7: a negative dot product means an error

Wrong: "the dot product must be positive". Right: negative is fine: it just means the angle is obtuse (Example 6b gives -4).

Misconception 8: "unit vector" means a vector with a 1 in it

Wrong: $(1,1)$ is a unit vector. Right: $\|(1,1)\| = \sqrt{2} \neq 1$. The unit vector is $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

16 More worked examples

Example: A vector of a given length

Find a vector of length 10 in the direction of $\mathbf{v} = (3, -4)$. $\|\mathbf{v}\| = 5$, unit vector $\left(\frac{3}{5}, -\frac{4}{5}\right)$, times 10: $(6, -8)$. Check: $\sqrt{36 + 64} = 10$ ✓.

Example: Is the triangle right-angled?

$A(1,1)$, $B(4,2)$, $C(2,4)$. $\overrightarrow{AB} = (3,1)$, $\overrightarrow{AC} = (1,3)$, $\overrightarrow{AB} \cdot \overrightarrow{AC} = 3 + 3 = 6 \neq 0$. $\overrightarrow{BA} = (-3,-1)$, $\overrightarrow{BC} = (-2,2)$: $6 - 2 = 4 \neq 0$. $\overrightarrow{CA} = (-1,-3)$, $\overrightarrow{CB} = (2,-2)$: $-2 + 6 = 4 \neq 0$. No right angle.

Example: Angle between $(1,2,2)$ and $(2,-1,2)$

Dot $= 2 - 2 + 4 = 4$; norms 3 and 3; $\cos\theta = \frac{4}{9}$; $\theta = \cos^{-1}(0.444) \approx 63.6°$.

17 One-page summary

Norm, dot product and distance in one page

- **Norm:** $\|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}$ (Pythagoras). $\|\mathbf{v}\| \geq 0$, $= 0$ only for $\mathbf{0}$, $\|k\mathbf{v}\| = |k|\|\mathbf{v}\|$.
- **Unit vector:** $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$. Standard: $\mathbf{i}, \mathbf{j}, \mathbf{k}$, $\mathbf{e}_1, \ldots, \mathbf{e}_n$; $\mathbf{v} = \sum v_i \mathbf{e}_i$.
- **Distance:** $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\sum (u_i - v_i)^2}$.
- **Dot product:** $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta = u_1 v_1 + \cdots + u_n v_n$ (a number). Shadow picture.
- **Angle:** $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$; > 0 acute, $= 0$ perpendicular, < 0 obtuse.
- $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$; symmetric, distributive, homogeneous, positive.
- **Cauchy–Schwarz:** $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$. **Triangle:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.
- **Parallelogram:** $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$; $\mathbf{u} \cdot \mathbf{v} = \frac{1}{4}\|\mathbf{u} + \mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{u} - \mathbf{v}\|^2$.
- **Matrices:** $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v}$; $A\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot A^T\mathbf{v}$; entries of AB are row·column.

18 Practice

Anton, Exercise Set 3.2: **1–8** (norms, unit vectors, $\|k\mathbf{v}\|$), **9–14** (dot products, distance, angles), **15–16** (does it make sense?), **17–18** (Cauchy–Schwarz), **19–20** (distance sets), **21–25** (direction angles, cube, box), **26–30** (thinking questions). Full solutions: *Norm_Dot_Distance_Solved_Exercises.pdf*.